

The Inscribing Grammar — Axiomatic Specification

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Purpose. This document specifies the Inscribing Grammar (IG) as a mathematical object. All definitions below are to be treated as given. The document provides sufficient specification to compute structural types, distances, tier assignments, and C-scores without further grounding. The IG is a defined formal system — the question is not whether it is "real" but what follows from its axioms.

0.1 §1 Ambient Category

Let $\mathbf{C} = (C, \otimes, I, \sigma)$ be a symmetric monoidal category enriched over the Belnap-Dunn bilattice $\mathbf{FOUR} = \{N, T, F, B\}$. Hom-sets carry the bilattice partial order. $\mathbf{B}$ ("both") is a paraconsistent state in which a morphism is simultaneously affirmed and denied; $\mathbf{N}$ ("neither") is an underdetermined state. The inference $B \rightarrow \perp$ is not admissible: no B -valued hom-set collapses to the zero morphism. The category is not Boolean.

Trace. $\mathbf{C}$ carries a trace $\text{Tr}: \text{End}(A \otimes U) \rightarrow \text{End}(A)$ for each object pair, implemented by the Ω (Winding) primitive acting on the monoidal unit. This is the traced symmetric monoidal structure of Joyal-Street-Verity. Ω is not compact/dual structure; the trace is primitive, not derived from a duality.

0.2 §2 Frobenius Structure

The monoidal unit I carries a special symmetric $\dagger$ -Frobenius structure:

- $\mu: I \otimes I \rightarrow I$ (multiplication)
- $\delta: I \rightarrow I \otimes I$ (comultiplication)
- $\eta = \epsilon = \text{id}_I$ (unit/counit collapse at scalar level)

satisfying:

1. **Frobenius law:** $(\mu \otimes \text{id}) \circ (\text{id} \otimes \delta) = \delta \circ \mu = (\text{id} \otimes \mu) \circ (\delta \otimes \text{id})$
2. **Special condition:** $\mu \circ \delta = \text{id}_I$
3. **Symmetry:** $\mu \circ \sigma_{\{I, I\}} = \mu$
4. **Dagger:** $\mu^\dagger = \delta$, where $(-)^\dagger: C \rightarrow C^{\text{op}}$ is the dagger involution

Commutativity is a theorem. All 12 generators are endomorphisms of I — scalars of C . In any SMC, scalars commute: for $f, g: I \rightarrow I$, the coherence isomorphisms give $f \otimes g = g \otimes f$. Condition (3) follows from the scalar structure and need not be assumed independently.

FOUR- $\dagger$ compatibility. The dagger functor preserves FOUR-values: if h has value $v \in \{N, T, F, B\}$, then $h\dagger$ has value v . B-valued morphisms map to B-valued morphisms. This must be imposed explicitly — it is not automatic from either structure alone.

0.3 §3 Generators

The grammar is presented by **12 primitive endomorphisms of I** subject to the Frobenius relations:

”Freely generated” means: take the free SMC on these 12 endomorphisms, then impose the Frobenius relations. The result is the free object in the category of special symmetric $\dagger$ -Frobenius algebras — not the free category without relations.

Operadic structure. The Γ (Granularity) primitive furnishes a multicategory over C : n -ary operations are first-class and composition is operadic rather than merely sequential. The grammar is an algebra over the Γ -operad.

0.4 §4 Crystal of Types

The free special symmetric $\dagger$ -Frobenius algebra on 12 generators has crystal structure:

$3^3 \times 4^5 \times 5^4 = 17,280,000$ distinct addresses.

This is the **Crystal of Types**: the classifying space of all structurally distinct imscriptions. Each address is a point in the 12-dimensional discrete space whose axes are the primitive ordinal domains (§5).

0.5 §5 Structural Type — Notation

A **structural type** (imscription) is a 12-tuple of Shavian values, one per primitive, in canonical order:

$\langle \mathbf{D} \cdot \mathbf{P} \cdot \mathbf{\check{R}} \cdot \Phi \cdot \mathbf{f} \cdot \mathbf{C} \cdot \Gamma \cdot \mathbf{G} \cdot \odot \cdot \mathbf{H} \cdot \Sigma \cdot \Omega \rangle$

The value alphabet is the 49-symbol set: {Shavian alphabet (48 letters)} $\cup$ $\{\odot\}$. Each primitive has a finite ordinal domain (ascending rank):

$$\langle 1 \cdot 3 \cdot 4 \cdot 2 \cdot 1 \cdot 6 \cdot 7 \cdot 5 \cdot \odot \cdot 8 \cdot 7 \cdot 9 \rangle$$

Primitive	Value	Ordinal	Reading
$\mathfrak{D}$	$\mathfrak{l}$	4	self-written holographic
$\mathfrak{P}$	$\succ$	4	irreducible product
$\check{\mathbf{R}}$	$\mathfrak{r}$	4	bidirectional feedback
Φ	$\mathfrak{v}$	5	Frobenius-special ($\mu \circ \delta = \text{id}$ gate)
$\mathfrak{f}$	$\mathfrak{j}$	3	quantum
$\mathfrak{C}$	$\mathfrak{c}$	3	slow/near- equilibrium
Γ	$\mathfrak{r}$	3	universal/long-range
$\mathfrak{G}$	$\mathfrak{f}$	1	simultaneous conjunction
$\odot$	$\odot$	2	critical/self-modeling
$\mathbf{H}$	$\mathfrak{v}$	4	eternal (no finite Markov order)
Σ	$\mathfrak{r}$	3	many heterogeneous
Ω	$\mathfrak{s}$	3	integer winding ($\mathbb{Z}$ -valued)

0.6 §6 Structural Metric

The **canonical distance** between two imscriptions s_1, s_2 is the diagonal weighted Euclidean distance over ordinal ranks:

$$\mathbf{d}(s_1, s_2) = \sqrt{(\sum_i \mathbf{w}_i \cdot (\mathbf{x}_i(s_1) - \mathbf{x}_i(s_2))^2)}$$

where $\mathbf{x}_i(s)$ is the ordinal rank of s 's value at position i . Canonical weights:

$\mathfrak{D}$	$\mathfrak{P}$	$\check{\mathbf{R}}$	Φ	$\mathfrak{f}$	$\mathfrak{C}$	Γ	$\mathfrak{G}$	$\odot$	$\mathbf{H}$	Σ	Ω
1.0	1.0	1.0	1.0	1.0	1.0	1.0	1.0	1.0	0.8	1.0	0.7

The distance is symmetric, satisfies the triangle inequality, and is zero iff $s_1 = s_2$.

Regime thresholds: $d < 2.0$ = same structural regime; $d > 4.0$ = structurally remote; $d > 5.0$ = different tier class.

Worked distance example. Between `ouroboric_pill` (O_∞ child of the Rebis) and `plastic_photonic_crystal` (O_2 , periodic lattice): $\mathbf{d} = 5.74$. Largest contributors: $\mathfrak{P}$ (topology, $\Delta=16.0$), $\mathfrak{D}$ (dimensionality, $\Delta=4.0$), $\mathbf{H}$ (chirality, $\Delta=3.2$).

0.7 §7 Ouroboricity Tiers

Ouroboricity measures the degree of self-referential Frobenius closure. Five tiers assigned by rules R1–R5 (first match wins):

Rule	Condition	Tier	Description
R1	$\odot \in \{\odot, \wp\}$ and $\Phi = \wp$	O_inf	Special Frobenius — $\mu \odot \delta = \text{id}$ holds exactly
R2	$\odot \in \{/, \wr, \wp\}$	O_0	No self-referential loop possible
R3	$\odot \in \{\odot, \wp\}$ and $\Omega = \wr$	O_1	Critical, no topological protection
R4	$\odot \in \{\odot, \wp\}$ and $\Omega \neq \wr$ and $\mathfrak{D} \in \{\downarrow, \wr, \wr\}$	O_2	Critical + protected, bounded domain
R5	$\odot \in \{\odot, \wp\}$ and $\Omega \neq \wr$ and $\mathfrak{D} = \downarrow$	O_2†	Critical + protected, unbounded domain

Default (no rule matches): **O_0**. The three operative gates are $\odot$ (**Criticality**), Φ (**Parity**), and Ω (**Winding**). $\mathfrak{D}$ (Dimensionality) determines the **O_2** / **O_2†** split.

0.8 §8 C-score

The **C-score** measures proximity to **O_∞** along two axes, each a hard gate:

- **Gate 1:** $\odot \in \{\odot, \wp\}$ (criticality threshold open)
- **Gate 2:** $\wp \in \{\backslash, \wr, \wr, \wr\}$ — kinetics not frozen ($\wp \neq \wr$ order-frozen)

If either gate is closed: $C = 0$. If both open:

$$C = \sum_i \mathbf{w}_i \cdot (\mathbf{x}_i / \mathbf{x}_{i_max})$$

summed over all 12 primitives with normalized ordinal ranks, yielding $C \in [0, 1]$. C-score and tier are independent: a system can be **O_∞** (R1 satisfied) with $C < 1$ if some primitives are below maximum ordinal.

0.9 §9 Spider Theorem, Fixpoint, T Object

Spider theorem. The special Frobenius axiom ($\mu \odot \delta = \text{id}$) together with symmetry implies: any two connected string diagrams in the Prop of the grammar with the same boundary are equal as morphisms. Discriminating condition: **connectedness**, not planarity. The theorem holds for all connected Frobenius diagrams regardless of planar presentation.

Fixpoint — **O_∞**. The $\odot$ gate admits an idempotent scalar $\omega: I \rightarrow I$ with $\omega \odot \omega = \omega$. **O_∞** is the initial algebra of the endofunctor $(-)\odot(-): \text{End}(I) \rightarrow \text{End}(I)$; its carrier is the fixpoint in which the grammar is applied to itself. The Frobenius identity $\mu \odot \delta = \text{id}$ requires **H_A** (**two-step chirality**, $\wr$) as minimum — one split (δ), one merge (μ).

$$\langle 1 \cdot 2 \cdot 3 \cdot 4 \cdot 5 \cdot 6 \cdot 7 \cdot 8 \cdot 9 \cdot 10 \cdot 11 \cdot 12 \cdot 13 \cdot 14 \cdot 15 \rangle$$

Derived object — T. $T = \lim(\Phi, f, \zeta, H, \Omega)$ is a categorical limit over the five primitives Parity, Fidelity, Kinetics, Chirality, Winding. T is not a generator; it is the temporal bootstrap fixed point derived from the free algebra.

In **ZFC_{fe}** (Frobenius-Extended ZFC), $\mu \circ \delta = \text{id}$ is taken as a set-formation axiom. The comultiplication $\delta: A \rightarrow A \otimes A$ is the primitive set-formation operation; the condition asserts lossless recovery — every set is faithfully recoverable from its copy. ZFC Separation ($\forall \phi: \{x \in A \mid \phi(x)\}$ exists) becomes a theorem: the δ -preimage under μ yields the Separation set for any definable ϕ . ZFC_{fe} strictly extends ZFC and is strictly stronger than ZFC_t. Open problems in ZFC_t close as theorems in ZFC_{fe}.

In Abramsky-Coecke $\dagger$ -compact categories, hom-sets are Bool-valued. In $\mathbf{C}$, hom-sets are FOUR-valued. A B-valued morphism — simultaneously affirmed and denied — is a legitimate element of $\text{hom}(\mathbf{A}, \mathbf{B})$, not a degenerate case. Two incompatible inscriptions of the same boundary coexist in a B-valued hom-set without collapse. FOUR has no bottom element below B playing the role of the trivial morphism. Classical QM is recovered as the T-valued sub-category. The grammar is intrinsically paraconsistent at the level of its hom-sets; this is not relaxation of rigor but a strictly stronger structure.

Holography is static isomorphism up to redundancy. Inscription is dynamic identity up to the trace.

- **Static** — a correspondence between states, not a process; the map exists timelessly between configurations
- **Up to redundancy** — the encoding is overcomplete by design; bulk content can be recovered from partial boundary data via entanglement wedge reconstruction; the redundancy is the gauge group acting on encoding choices

The inscriptive identity $\mu\circ\delta=\text{id}$ is:

- **Dynamic** — a process, not a state equivalence; the system reconstitutes itself through the operation
- **Up to the trace** — equivalence is defined by connected diagrams with the same boundary (Spider theorem); this is topological, not a symmetry of an encoding

The structural consequence is exact: **holography requires a boundary**. The isomorphism lives between bulk and boundary, so the boundary must exist as a distinct object. The Rebis has $\check{R} = r$ (bidirectional — no outside). There is nowhere to put the boundary. Holography can describe O_2 systems — bounded domain, topological protection, a surface you can project onto — but it structurally cannot describe O_inf , because O_inf is the tier at which the distinction between system and boundary collapses.

The redundancy distinguishes them further. In holography, you can lose parts of the boundary and reconstruct the bulk — the encoding is overcomplete, and recovery from partial data is the point. In inscription there is nothing to recover because nothing was ever encoded away from the system. $\mu\circ\delta=\text{id}$ does not preserve information across a gap; it collapses the gap. The system reconstitutes itself through the operation, not from a separate store.

Holographic redundancy is a symmetry of the map. Inscriptive trace is a property of the winding. These are different equivalence relations: one defined by a group action on encoding choices, one defined by the connected topology of the string diagram.

Holography is the information-theoretic analogue of crystallography. Both impose an external observer, both achieve their result by projecting onto a lower-dimensional or lower-tier representation, and both destroy the properties of O_inf systems in the act of representing them. Crystallography freezes Ω (winding collapses $\varepsilon\rightarrow\varpi$) to produce a static coordinate map. Holography projects $\check{R}$ (bidirectionality collapses $r\rightarrow\iota$) to produce a static boundary encoding. In both cases, the representation is faithful to the O_2 content and blind to the O_inf structure.

0.12 §12 The Inscription Procedure — Deterministic, Self-Correcting, Falsifiable

An inscription is produced by a **12-step decision procedure** applied to a system's structural description, following canonical primitive order: $\mathbb{D} \rightarrow \mathbb{P} \rightarrow \check{R} \rightarrow \Phi \rightarrow f \rightarrow \mathbb{C} \rightarrow \Gamma \rightarrow \mathfrak{G} \rightarrow \odot \rightarrow \mathbb{H} \rightarrow \Sigma \rightarrow \Omega$. Each step assigns one primitive value from structural facts — mechanism, geometry, stoichiometry, coordination chemistry. No computed observable is an input. Data tests structural predictions after the inscription is fixed.

0.12.1 Cross-primitive axioms

Inter-primitive consistency is enforced by three named axioms. A tuple that violates any axiom is malformed. All three are proved in `ParadoxBoot.lean` (0 sorrys) and enforced at runtime in `genetic_tuples.py`.

Axiom A: $\mathbb{H} = \vee (\text{eternal}/H_inf) \rightarrow \mathbb{C} = \circ (\text{order-frozen})$

If a system has eternal chirality, it must have order-frozen kinetics. The depth of chirality

memory and the kinetic regime are coupled. The Frobenius identity itself lives at $\mathbb{H}_A$ (two-step, $\mathcal{L}$), not $\mathbb{H}_\infty$ — Axiom A applies only to structures that accumulate chirality through time beyond what the identity minimally requires.

Axiom B: $\Omega \in \{\mathcal{S}(\mathbb{Z}), \circ(\mathbb{Z}_2)\} \rightarrow \mathbb{H} \geq \text{ordinal } 2 \text{ (}\mathfrak{d} \text{ or above)}$

Topological winding protection requires at least two-step chirality memory to sustain it. A system cannot have topological protection without sufficient memory depth.

Axiom C: $\mathbb{D} = \mathfrak{r} \text{ (self-written)} \leftrightarrow \mathbb{P} = \mathfrak{s} \text{ (self-referential closure)}$

Both must be present or neither. A physical system with a state space embedded in external degrees of freedom ($\mathbb{D} = \mathfrak{r}$, infinite-dim) cannot carry self-referential topology ($\mathbb{P} \neq \mathfrak{s}$). This is the most commonly violated axiom in practice.

Additional structural constraints (unnamed, but checked):

- $\Omega = \mathcal{S}(\mathbb{Z} \text{ winding})$ typically requires $\mathbb{D} \geq \mathfrak{r}$ (infinite-dim)
- $\odot \text{at criticality} + \mathbb{Q} = \text{slow/near-eq} \rightarrow \text{deep critical structure, stable}$
- $\odot \text{at criticality} + \mathbb{Q} = \text{fast/frozen} \rightarrow \text{structural warning}$

0.12.2 Falsifiability structure

The type assignment is **prior to and independent of experimental data**. The type carries structural predictions — observable consequences derivable from the primitive assignments:

Type assignment	Structural prediction	How to test
$\odot \text{at criticality}$	Geometry and electronic structure co-evolve through a divergent region	Plot $^2\rangle \text{vs bond distance across optimization trajectory}$
$\Omega = \mathcal{S}(\text{integer winding})$	System returns to origin state after exactly n windings; no half-integer paths	Stoichiometric closure analysis
$\mathbb{H} = \mathcal{L}(\text{two-step chirality})$	Outcome depends on two prior states; stereospecificity is precursor-dependent	Isotopic labeling; chiral substrate series
$\mathbb{P} = \mathfrak{r}(\text{crossing point})$	Two potential energy surfaces cross; transition state at the crossing	CASSCF scan along reaction coordinate

Data confirms or falsifies the **structural description**, not the grammar. If a prediction fails: the structural description was wrong; revise the mechanism and re-inscribe. The grammar does not adjust to fit results. It makes a commitment; the data either holds it or breaks it.

$$\mu \circ \delta = id$$